Akhlaque Ahmad

AI Full Stack Engineer – Angular & Node.js Architecture – AI/Agentic Systems

akhlaquea01@gmail.com | +91-8797247279 | Bengaluru, India |Linkedin | Github | Akhlaque

Professional Summary

Senior Software Engineer with 5.5+ years of experience developing enterprise B2B SaaS applications using Angular, TypeScript, JavaScript, Node.js, Express.js, MongoDB, and REST APIs. Specialized in scalable frontend architecture, reusable component development, Node.js BFF services, and AI-enabled enterprise applications. Hands-on experience integrating LiteLLM/OpenAI models, Server-Sent Events (SSE), MCP, and Web Components to build intelligent sourcing solutions. Experienced in performance optimization, production support, code reviews, feature ownership, and delivering high-quality software in Agile environments.

Technical Skills

- **AI & Enterprise Integration:** LiteLLM, OpenAI Models, MCP (Model Context Protocol), Server-Sent Events (SSE), AI Chat Integration, Agentic Workflows, AI Workflow Visualization, Data Ingestion Pipelines
- **Programming Languages:** TypeScript, JavaScript (ES6+), HTML5, CSS3, SCSS, SQL
- **Backend:** Node.js, Express.js, REST APIs, Backend-for-Frontend (BFF), Swagger/OpenAPI, Joi Validation, Event-Driven Architecture, MongoDB CRUD Operations
- **Frontend:** Angular (11–18), RxJS, Angular Signals, Standalone Components, Angular Material, Angular CDK, Web Components, React Web Component Integration, Micro Frontends, Custom NPM Packages, Ably Pub/Sub, GoJS, Highcharts
- **Databases:** MongoDB, MySQL, PostgreSQL (Basic)
- **Tools & Platforms:** Git, GitLab, Jenkins, Docker, AWS, SonarQube, Consul, HashiCorp Vault
- **Monitoring & Debugging:** Datadog, Dynatrace, Elastic APM, Rookout
- **Performance & Quality:** Lazy Loading, Angular Signals, OnPush Change Detection, Virtual Scrolling, TrackBy, Code Splitting, Bundle Optimization, Accessibility (WCAG), Jest, Karma
- **Development Practices:** Agile Scrum, Code Reviews, Feature Ownership, Production Support, Release Management

Work Experience

Senior Software Engineer – *Zycus Infotech, Bengaluru, India* Feb 2021 – Present

- Architected enterprise B2B SaaS procurement platforms (iSource, BOM/BOQ) using Angular, Node.js BFF microservices, MongoDB, and serving multi-tenant buyer and supplier personas.
- Designed reusable BFF aggregation layers and API contracts across 5+ Node.js microservices, standardizing enterprise consumption and reducing frontend integration effort by 30%.
- Engineered lazy-loaded Angular architectures (10+ modules) and optimized virtual-scrolled grids for 10,000+ row pricing tables, boosting page performance by 40% via Angular Signals and Standalone Components.
- Automated workflows and built async Excel import/export pipelines handling 500MB+ uploads via job queues and worker threads, decreasing manual procurement effort by 50%.
- Integrated OpenAI APIs, LiteLLM, and MCP servers with SSE-based streaming to develop an autonomous negotiation chat system for AI-assisted sourcing.
- Elevated deployment reliability by establishing multi-gate pre-PR validations (accessibility, linting, unit testing, SonarQube).
- Mentored 3+ engineers through code reviews and architecture discussions while collaborating closely with Product Managers, QA, and Backend Architects in Agile delivery environments.

Key Achievements

- Reduced manual sourcing effort by 50% through advanced workflow automation and optimization.
- Increased application performance by 40% by spearheading frontend modernization initiatives.
- Recognized as Employee of the Month four times for technical leadership and consistent delivery excellence.

Projects

iSource – Enterprise Strategic Sourcing Platform

Technologies: Angular 14, TypeScript, RxJS, Node.js, Express, Spring Boot, PostgreSQL, Redis, AWS S3, MCP, SSE

- Architected a high-scale, three-tier multi-tenant sourcing platform (Angular SPA → Node.js BFF → Spring Boot) featuring a lazy-loaded architecture of 30+ feature and 10+ routing modules.

- Built a dual-shell routing system for diverse user personas, utilizing route guards and resolvers for strict activity-based authorization and efficient data pre-fetching.
- Developed dynamic questionnaire engines, real-time multilingual auction monitoring, and Analysis 360 award-scenario comparison tools utilizing Highcharts.
- Delivered an AI-powered autonomous negotiation chat featuring SSE streaming, multi-agent delegation, and MCP servers that exposed BFF endpoints as callable tools.
- Designed a Node.js Express BFF leveraging the API Quad pattern across 30+ entity groups, implementing SSO authentication, Joi validation, and Consul service discovery.

BOM/BOQ Platform – Procurement Pricing Micro-Application
Technologies: Angular 11, Node.js, Express, MongoDB, AWS S3, mathjs, Docker/ECR

- Designed a multi-service Bill of Materials platform as an Angular micro-frontend embedded in a host shell, orchestrated by a primary BFF gateway and four specialized Node.js microservices.
- Implemented high-performance pricing grid components utilizing CDK virtual scroll and debounced inputs to flawlessly render base, alternate, and additional item data.
- Created a drag-and-drop BOM template builder with 15+ customizable column types and an advanced mathjs-based formula engine for dynamic calculated pricing.
- Engineered an asynchronous Excel processing pipeline to handle massive datasets, utilizing chunked API persistence to eliminate timeout failures.
- Secured multi-tenant deployments by integrating HashiCorp Vault, Redis session caching, and JWT middleware, alongside containerizing all 6 services with Docker.

Merlin – Agentic AI Workflow Platform
Technologies: Node.js, Angular, LiteLLM, GoJS, MongoDB, REST APIs

- Developed the comprehensive frontend and integration layer for a procurement automation platform, bridging intuitive UIs with a robust external agentic workflow engine.
- Engineered interactive visual node-graph workflows with GoJS to streamline supplier discovery and business process mapping.
- Built sophisticated data ingestion pipelines tailored to structure complex procurement data for optimal LLM consumption and AI-assisted event generation.

Education

Bachelor of Technology (B.Tech.), Computer Science Engineering – *Vishveshwarya Group of Institutions (Affiliated with AKTU), Ghaziabad, India* 2016 – 2020

Certifications

AI-First Software Engineering Program – *Airtribe* Ongoing

Awards & Recognitions

- Employee of the Month – Apr 2022, Apr 2023, Nov 2024, Apr 2025.
- Consistently recognized by leadership for innovation, technical excellence, and rapid project delivery.